

I. Classification Loss Functions (1–10)

1. Q: For a multiclass classification problem, what is the most commonly employed loss function for training a decision tree?

A: **Multinomial Cross-Entropy Loss** (also called Softmax Loss).

2. Q: What is the primary goal of minimizing the multinomial cross-entropy loss?

A: To maximize the likelihood of correctly predicting the true class labels and reduce misclassification.

3. Q: Write the full formula for multinomial cross-entropy loss (L_{softmax}) for (n) datapoints and (C) classes.

A:

$$L_{\text{softmax}} = -\sum_{i=1}^n \sum_{c=1}^C I(y_i = c) \log(\hat{P}_{ic})$$

4. Q: In the cross-entropy loss formula, what does $(\hat{P}_{ic})$ represent?

A: The predicted probability that datapoint (i) belongs to class (c).

5. Q: What does the indicator function $(I(y_i = c))$ contribute to the total loss?

A: It contributes **only when the true class of (i) is (c)**, i.e., it selects the relevant log probability.

6. Q: Minimizing cross-entropy subject to $(\sum_{c=1}^C \hat{p}_{kc} = 1)$ is solved using what technique?

A: **Lagrange Multipliers.**

7. Q: Formula for optimal prediction probability $(\hat{p}_{kc})$ *minimizing the Lagrangian*?

A:

$$\hat{p}_{kc} = \frac{1}{|R_k|} \sum_{\vec{x}_i \in R_k} I(y_i = c)$$

8. Q: The optimal $(\hat{p}_{kc})$ represents the fraction of points in (R_k) belonging to which class?

A: Class (c).

9. Q: Regression trees assign the leaf prediction $(\hat{c}_k)$ *as the average; what is the equivalent in classification trees*?

A: The **class probabilities** $(\hat{p}_{kc})$ for each class (c).

10. Q: For binary classification, what does multinomial cross-entropy reduce to?

A: **Binary Log Loss (Binary Cross-Entropy Loss).**

II. Tree Splitting and Impurity Measures (11–20)

11. Q: What two criteria were used for recursive binary splitting in regression trees?

A: **Sum of Squared Errors (SSE)** reduction and variance minimization.

12. Q: For classification trees, recursive binary splitting attempts to maximize the reduction of what?

A: **Node purity**, i.e., **reduction in impurity**.

13. Q: What is required for a split to be considered “good”?

A: The child nodes must be **more homogeneous** (lower impurity) than the parent node.

14. Q: What are the two key measures of node impurity?

A: **Gini Impurity** and **Entropy**.

15. Q: Formula for Gini Impurity ($G(R_k)$) for (d) classes?

A:

$$\begin{aligned} &[\\ G(R_k) &= 1 - \sum_{c=1}^d \hat{p}_{kc}^2 \\ &] \end{aligned}$$

16. Q: If (R_k) is perfectly pure, what is ($G(R_k)$)?

A: ($G(R_k) = 0$).

17. Q: For a binary class region with equal points from both classes, what is the Gini Impurity?

A: ($G(R_k) = 0.5$) (maximum impurity).

18. Q: Formula for Entropy ($H(R_k)$) for (d) classes?

A:

$$\begin{aligned} &[\\ H(R_k) &= -\sum_{c=1}^d \hat{p}_{kc} \log_2 (\hat{p}_{kc}) \\ &] \end{aligned}$$

19. Q: If a region is perfectly pure, what is ($H(R_k)$)?

A: ($H(R_k) = 0$).

20. Q: What is the formal name for the largest reduction in impurity from a split?

A: **Information Gain** (or Impurity Reduction).

III. Prediction and Structure (21–30)

21. Q: How is the final prediction for a datapoint in a region (R_k) determined?

A: By choosing the **class with the highest probability** in that region (majority class).

22. Q: How does the final prediction in classification differ from regression?

A: Regression outputs a **continuous value** (mean), classification outputs a **discrete class label or probability vector**.

23. Q: What does (m) represent in the prediction formulas?

A: The **number of leaf nodes** in the final decision tree.

24. Q: Formula for Information Gain after a split?

A:

$$\begin{aligned} & [\\ & \text{IG} = I(R_{\text{parent}}) - \left(\frac{|R_{\text{left}}|}{|R_{\text{parent}}|} I(R_{\text{left}}) + \right. \\ & \left. \frac{|R_{\text{right}}|}{|R_{\text{parent}}|} I(R_{\text{right}}) \right) \\ &] \end{aligned}$$

25. Q: What must be true about child node impurity for a split to be successful?

A: The **weighted impurity** of child nodes must be **less than the parent node impurity**.

26. Q: How are left and right child nodes weighted in Information Gain calculation?

A: By the **fraction of datapoints** in each child relative to the parent node.

27. Q: Why is homogeneity a key focus in classification splits?

A: It ensures that each leaf node represents a **predictable and uniform class**, improving accuracy.

28. Q: To prevent overfitting, after growing a tree, what process is applied?

A: **Pruning** (reduces tree size and improves generalization).

29. Q: In the prediction formula, what is the role of the Lagrange multiplier (λ)?

A: Ensures that the **sum of predicted probabilities equals 1**.

30. Q: What is the relationship between Log Loss and the optimal probabilities ($\hat{p}_{kc}$) in leaves?

A: *Minimizing Log Loss produces **optimal class probability estimates** ($\hat{p}_{kc}$) for each leaf.*